Summary in Graph

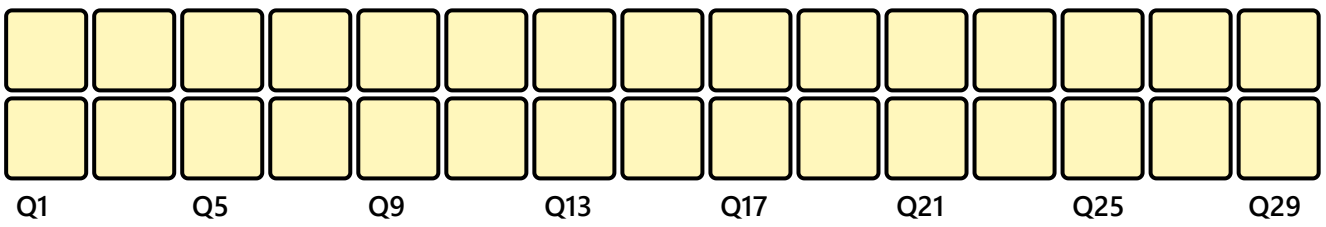
# Exam Summary (GATE Overflow CS Test Series 2026 | Mix Subjects | Test 4)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 30 (10 + 20) |
| Total Marks: | 50 (10 + 40) |
| Exam Duration: | 90 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (2) out of 19 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Computer Networks

Mark all the correct pairs in the following pairs of OSI protocol layer/sub-layer and its functionality. (Mark all the appropriate choices)

A. Transport layer and Routing
B. Physical Layer and Bit synchronization
C. Network layer and end-to-end connectivity

D. Medium Access Control sub-layer and Channel sharing

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)? (Mark all the appropriate choices)

A. SMTP, POP3
B. POP3, SMTP
C. SMTP, IMAP
D. SMTP, MIME

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are example(s) of stateless application layer protocol? (Mark all the appropriate choices)

A. POP3
B. HTTP
C. IMAP
D. UDP

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 03sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

The baud rate of classic $20$ Mbps Ethernet (in megabaud) is _____

Your Answer: | Correct Answer: 40 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Given an orthogonal matrix $A$, $(AA^T)^{-1}$ is _____

A. $2I$
B. $I$
C. $I/2$
D. $A$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

If $A = \begin{bmatrix} -12 & 7 \\ -5 & 3 \end{bmatrix}$, then the value of $\det(A^{2021})$ is _____

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #7** Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

If $\det \begin{pmatrix} 2 & \alpha \\ 3 & \beta \end{pmatrix} = 16$ and $\det \begin{pmatrix} 2 & \beta \\ 3 & \alpha \end{pmatrix} = 1$, then the value of $\alpha^2 + \beta^2$ is _____

Your Answer: | Correct Answer: 149 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #8** Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following tasks does a router $X$ in a packet-switched network perform when it gets a packet with destination address $d$? (Mark all the appropriate choices)

A. $X$ looks up $d$ in its routing table to determine the outgoing link.
B. $X$ calculates the minimum-cost route to destination $d$.
C. $X$ sends out a HELLO packet or a routing protocol advertisement to its neighbors.
D. $X$ may discard the packet without forwarding it.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #9** Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following properties of an application makes UDP transport layer protocol the recommended one? (Mark all the appropriate choices)

A. Speed
B. Flooding
C. Fairness between competing flows
D. Short duration interactions

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #10** Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are CORRECT? (Mark all the appropriate choices)

A. A repeater operates at the physical layer
B. A hub operates at the physical layer
C. A switch operates at the data-link layer
D. A bridge operates at the data-link layer

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Which of the following are possible with regard to processes in OS? (Mark all the appropriate choices)

A. A child process running itself even when its parent process has finished execution.
B. A process having an entry in the process table even after its execution being completed.
C. A process changing its parent using appropriate system calls.
D. A process having multiple parent processes at any time.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A path in a digital circuit switched network has a data rate of $2$ Mbps. The exchange of $2000$ bits is required for the setup and teardown phases. The distance between two parties is $10000$ km. If the propagation speed is $2 \times 10^8$ m/s, the total delay in milliseconds if $20000$ bits of data are exchanged during the data transfer phase will be _____

Your Answer: | Correct Answer: 213 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

In a packet switched network, packets are routed from source to destination along a single path having $3$ intermediate nodes. If the message size is $24$ bytes and each packet contains a header of $8$ bytes, then the optimum packet size in bytes is _____

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Two hosts are connected via a packet switch with $2 \times 10^7$ bits per second links. Each link has a propagation delay of $30$ microseconds. The switch begins forwarding a packet $40$ microseconds after it receives the same. If $13000$ bits of data are to be transmitted between the two hosts using a packet size of $6000$ bits. The time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is _____

Your Answer: | Correct Answer: 1050 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the set of processes whose arrival time and burst time are given below-

| Process Name | Arrival Time | Execution Time |
|---|---|---|
| $P_1$ | 5 | 5 |
| $P_2$ | 4 | 6 |
| $P_3$ | 3 | 7 |
| $P_4$ | 1 | 9 |
| $P_5$ | 2 | 2 |
| $P_6$ | 6 | 3 |

If the CPU scheduling policy is Round Robin with time quantum $= 3$ unit. The absolute value of the difference between average waiting time and average turn around time is (rounded to $2$ decimal places) _____

Your Answer: | Correct Answer: 5.33 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #16** Numerical Type | Award: 2 | Penalty: 0 | Operating System

Four processes arrive on a single processor at time $0$ in the order $A, B, C, D$. Their CPU bursts are $5, 2, 9, 2$ time units respectively. The completion time of $C$ under round robin scheduling with time slice of one time unit is _____

Your Answer: | Correct Answer: 18 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #17** Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following set of processes, assumed to have arrived at time $0$. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order $P_1, P_2, P_3, P_4$.

| Processes | $P_1$ | $P_2$ | $P_3$ | $P_4$ |
|---|---|---|---|---|
| Burst time (in ms) | 10 | 9 | 4 | 6 |

If the time quantum for RR is $5$ ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to $2$ decimal places is _____

Your Answer: | Correct Answer: 8.25 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #18** Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the set of process with arrival time (in milliseconds), CPU burst time (in milliseconds) and priority (higher value denoting higher priority) shown below. None of the processes have I/O bursts.

| Process | Arrival Time | Burst Time | Priority |
|---|---|---|---|
| $A$ | 0 | 4 | 2 |
| $B$ | 1 | 2 | 4 |
| $C$ | 2 | 3 | 6 |
| $D$ | 3 | 5 | 10 |
| $E$ | 4 | 1 | 8 |
| $F$ | 5 | 4 | 12 |
| $G$ | 6 | 6 | 9 |

If the average waiting time (in milliseconds) of all the process using non-preemptive priority scheduling algorithm is $'a'$ and the average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is $'b'$, then the value of $a + b$ (rounded to $1$ decimal point) is _____

Your Answer: | Correct Answer: 20.9:21 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #19** Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If $\alpha, \beta$, and $\gamma$ are the roots of the equation $x^3 + 3x^2 - 2x + 4 = 0$ then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is _____

Your Answer: | Correct Answer: 45 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following system of equations

$2x_1 + 3x_2 + 5x_3 + 7x_4 = 0$

$11x_1 + 13x_2 + 17x_3 + 19x_4 = 0$

$23x_1 + 29x_2 + 31x_3 + 37x_4 = 0$

$41x_1 + 43x_2 + 47x_3 + 53x_4 = 0$

The system has _____ (Mark all the appropriate choices)

A. a trivial solution
B. no solution
C. a non-trivial solution
D. a unique solution

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the $2 \times 2$ matrix, $A = \begin{bmatrix} 3 & 1 \\ -1 & 5 \end{bmatrix}$. What is the trace of $A^{20}$+ trace of $A^{21}$? (Mark all the appropriate choices)

A. $5 \times 2^{41}$
B. $2^{20} + 2^{21}$
C. $2^{40} + 2^{42}$
D. $2^{41} + 2^{43}$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

For the Matrix $A = \begin{bmatrix} 7 & 5 \\ 5 & 7 \end{bmatrix}$, the normalised Eigen Vector is/are _____. (Mark all the appropriate choices)

A. $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$
B. $\begin{bmatrix} \frac{-1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$
C. $\begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$
D. $\begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #23** Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following is correct?(Mark all the appropriate choices)

A. Open Shortest Path First (OSPF) is a unicast routing protocol.
B. In Link State protocols shortest path computations require many CPU circles.
C. No split horizon techniques are possible in the link state routing protocol.
D. Open Shortest Path First (OSPF) is a classless routing protocol.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #24** Numerical Type | Award: 2 | Penalty: 0 | Operating System

The arrival time, priority (lower value denoting higher priority), and duration of the CPU and I/O bursts for each of three processes $P_1, P_2$ and $P_3$ are given in the table below. Each process has a CPU burst followed by an I/O burst followed by another CPU burst. Assume that each process has its own I/O resource and hence do not block each other for I/O.

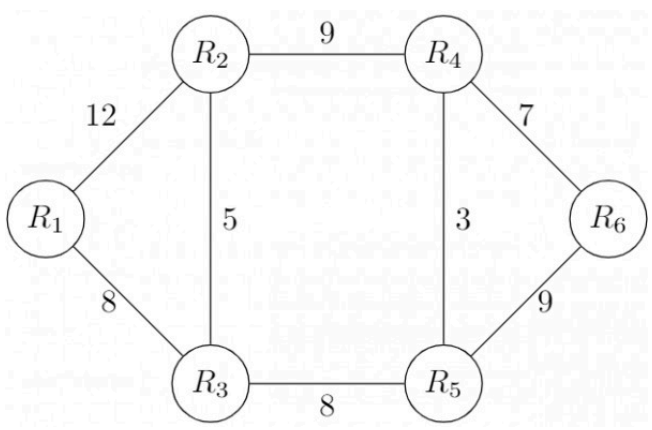
| Process | Arrival Time | Priority | Burst duration (CPU) | Burst duration (I/O) | Burst duration) (CPU) |
|---|---|---|---|---|---|
| $P_1$ | 0 | 2 | 3 | 7 | 5 |
| $P_2$ | 1 | 3 | 5 | 5 | 3 |
| $P_3$ | 3 | 1 | 4 | 5 | 3 |

If the multi-programmed operating system uses preemptive priority scheduling and the finish times of the processes $P_1, P_2$ and $P_3$ are $a, b$ and $c$ respectively, then $a + 2b + 3c =$?

Your Answer: | Correct Answer: 119 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #25** Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a network with $6$ routers $R_1$ to $R_6$ connected with links having weights as shown in the following diagram.

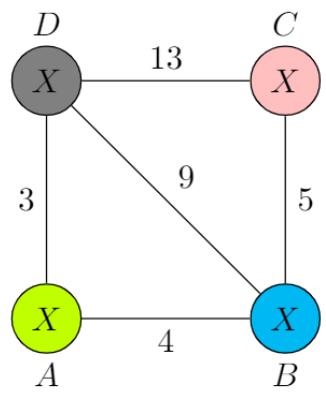



All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, the number of links in the network that will never be used for carrying any data $=$ _____

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #26** Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

There is a network consisting of $4$ routers, the weights (distance) being mentioned on the edges. Which of the following routing tables is/are correct? (Mark all the appropriate choices)




A.

Routing Table of D

| Destination | Distance | Next Hop |
|---|---|---|
| A | 3 | $A$ |
| B | 7 | $A$ |
| C | 12 | $A$ |
| D | 0 | $D$ |

B.

Routing Table of B

| Destination | Distance | Next Hop |
|---|---|---|
| A | 4 | $A$ |
| B | 0 | $B$ |
| C | 5 | $C$ |
| D | 7 | $A$ |

C.

Routing Table of A

| Destination | Distance | Next Hop |
|---|---|---|
| A | 0 | $A$ |
| B | 4 | $B$ |
| C | 9 | $B$ |
| D | 3 | $D$ |

D.

Routing Table of C

| Destination | Distance | Next Hop |
|---|---|---|
| A | 9 | $B$ |
| B | 5 | $B$ |
| C | 0 | $C$ |
| D | 13 | $D$ |

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #27** Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A desktop, a printer and a radio receptor are connected by a single wire to form a network capable of transmitting up to $4\ \mathrm{KBps}$. For a $5$ second period, all three devices were backlogged (i.e. had data to send). In these $5$ seconds, the desktop and the printer sent $8\ \mathrm{KB}$ each, while the radio receptor did not manage to send anything. If $U$ is the utilization $(U)$ and $F$ the fairness $(F)$ in the network during this period, the value of $U + F =$ _____

Your Answer: | Correct Answer: 1.3 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #28** Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following statements is/are true with regard to circuit switching and packet switching? (Mark all the appropriate choices)

A. Switches in a packet-switched network process connection establishment and tear-down messages, whereas switches in a circuit-switched network do not.
B. Under some circumstances, a circuit-switched network may prevent some senders from starting new conversations.
C. Unlike in packet switching, switches in circuit-switched networks do not need any information about the network topology to function correctly.
D. Once a connection is correctly established, a switch in a circuit-switched network can forward data correctly without requiring data frames to include a destination address.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following is correct? (Mark all the appropriate choices)

A. Through Address Resolution Protocol (ARP), $(32-\text{bit})$ IP address is mapped into $(48-\text{bit})$ MAC address.
B. In ARP, broadcast MAC address is used.
C. ARP is used in $IPv4$ and $IPv6$.
D. If the destination IP address is on a remote subnet, then ARP is not used.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following is correct?(Mark all the appropriate choices)

A. Internet Control Message Protocol (ICMP) can operate between host to host or host to router or router to router.
B. ICMP is a data-link layer (or) layer-$2$ protocol.
C. Internet group message protocol (IGMP) can be used between client and multicast router.
D. IGMP is a network layer (or) layer -$3$ protocol.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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